

Balance Sheet-Based Network Analysis of Indian Banking System

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Abstract: Banking systems are complex, dynamic networks that are typically robust to failure. The failure of a single institution, or slippages across several institutions, can lead to cascading failures in other banks due to inter-institution exposures. In this project, we have modeled the Indian banking system as a directed graph, with the edges representing interbank exposure. The adjacency matrix in this case is constructed using balance sheet data obtained from the annual reports of the respective banks, with the exact interbank exposure values being determined using the weights of the banks in the BANKNIFTY index. We have constructed thorough closed-loop mechanisms for financial shocks, which are used to perturb the system and perform contagion analysis, and have validated the logic of these shocks using the real-life case studies of: (i) Punjab National Bank crisis of 2017-18, (ii) YES Bank crisis of 2019–20.

Keywords: banking system, network analysis, balance sheet

1. INTRODUCTION

Financial systems are messy, complex networks where institutions interact with each other across different scales of volume. However, economic theory suggests that banking system failures and the resulting cascades are, to a large extent, driven by the interbank exposure network and the exposure of these banks to different markets [1-3].

However, we face data constraints on both fronts. We lack clear data on the exact interbank exposures and on the exposure of banks in different markets. If we assume that most banks hold diversified portfolios across markets (that is, they are not susceptible to major slippages unless global market-wide cascades occur), then the primary focus shifts to determining the interbank exposure network [1].

To determine the total advances a bank has with other banks, we look at Schedule 9 of the balance sheet (standalone financial statements), where we combine syndicated loans and advances to other banks in India. We then calculate the interbank exposure values using the weights of the banks in the BANKNIFTY index.

Though, this does not give the exact value of interbank exposures (in the case of banks such as State Bank of India, a significant portion of syndicated loans may be between subsidiary or group entities, which may not accurately reflect true interbank exposures) it serves as a good enough estimate for our purpose. Following this, we construct the banking network using objects/classes with the attributes shown in Fig. 1, (these attributes are derived from the respective schedules in the financial report).

There are two types of initiation shocks: risk transfer shocks and credit shocks. A risk transfer shock arises when a bank has made risky or fraudulent advances, as seen in the Punjab National Bank fraud case 2016-17. A credit shock arises from broader market crashes, where the

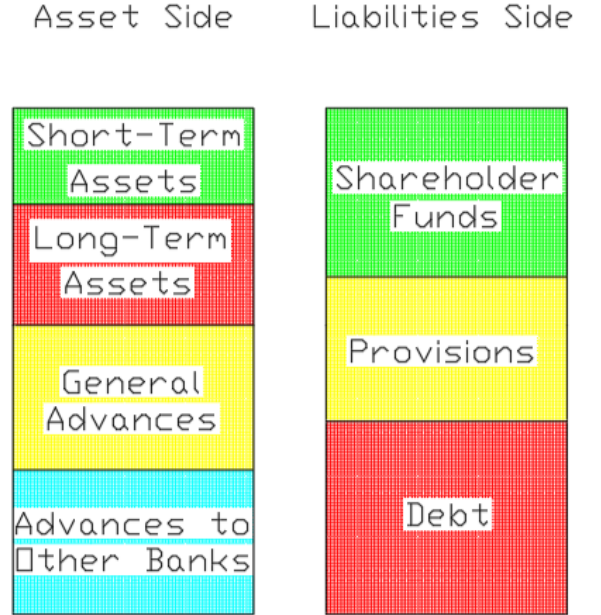


Fig. 1. Bank Object Structure

value of short-term assets such as gold or foreign currency declines due to global disruptions, as observed during the COVID-19 pandemic, which coincided with the YES Bank crisis of 2019–20. Both of these cases are illustrated in the toy examples in Figs. 2 (a) and (b), respectively (It is important to note that a risk transfer shock is initiated at a single bank, whereas a credit shock affects all banks in the network at the outset).

The third type of financial shock is the funding shock, which is used as a means of propagating the initial shock across the network over multiple iterations, and is illustrated in the toy example in Fig. 2 (c).

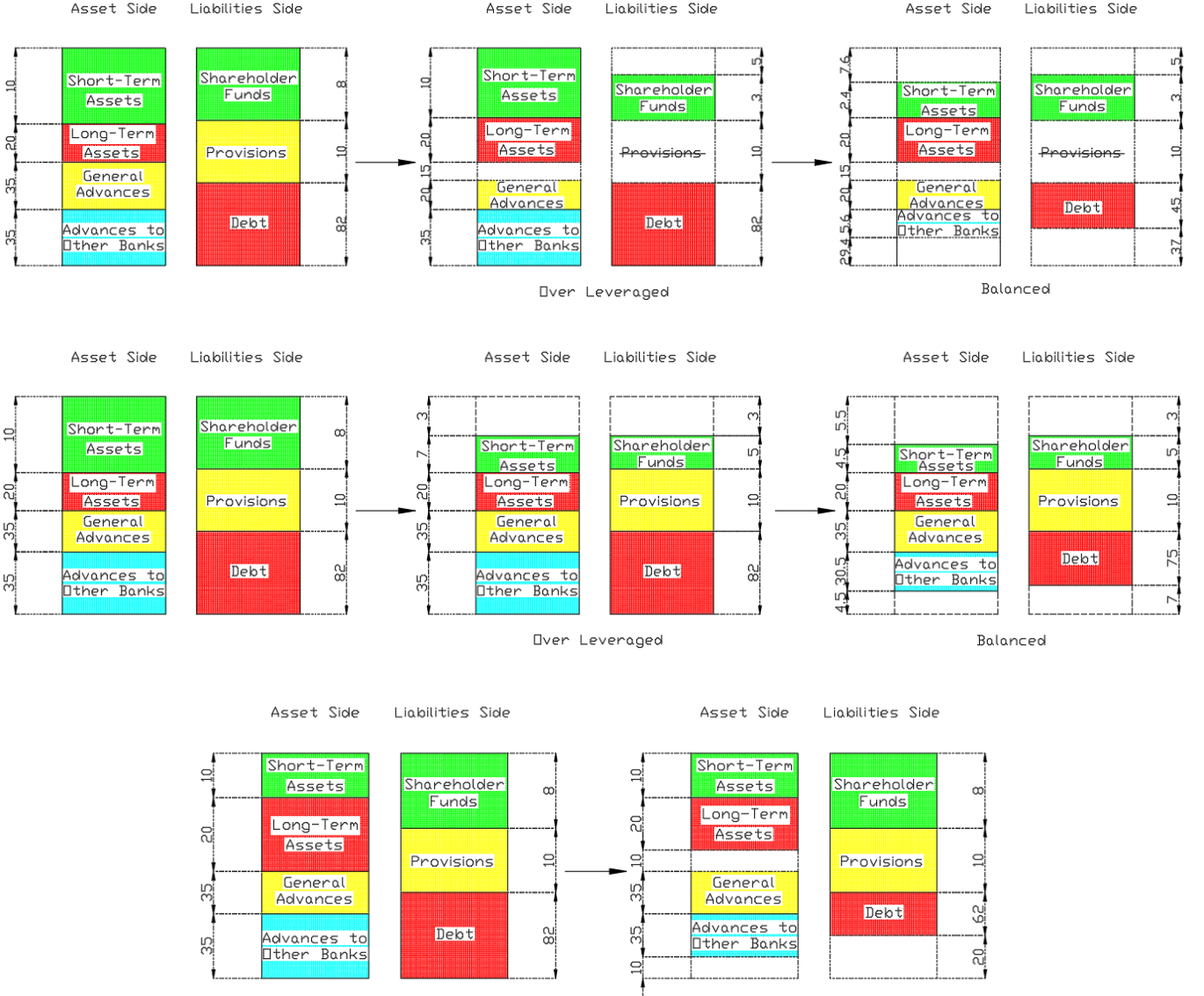


Fig. 2: Bank response to: (a) Risk transfer shock (Top), (b) Credit shock (Middle), and (c) Funding shock (Bottom).

2. METHODOLOGY

For a clearer understanding of the abbreviated terms used in the section below, refer to Appendix A: ONTOLOGY.

2.1 Setting up the Interbank Exposure Network

- (1) Each bank in the network is represented as a node, where a directed edge from Bank A to Bank B denotes an advance from Bank A to Bank B.
- (2) To distribute the total advances made by Bank A to other banks, the BANKNIFTY weights are first normalized after excluding Bank A, and these normalized weights are then used to allocate the advances across the remaining banks. This results in the fully connected network (Similar to Figs.3,4)
- (3) This is mathematically represented using the equations below:

$$r_{ij} = \frac{\alpha_j}{100 - \alpha_i}, \quad i \neq j \quad (2.1.1)$$

$$r_{ii} = 0 \quad (2.1.2)$$

$$w_{ij} = \frac{\alpha_j}{100 - \alpha_i} \cdot ATB_i, \quad i \neq j \quad (2.1.3)$$

- (4) In the above equations, α_j denotes the weight of a bank j in the BANKNIFTY index, r_{ij} represents the fraction of advances made by bank i to bank j, and w_{ij} represents the actual value of advances extended by bank i to bank j.

2.2 Defining Regulatory Ratios

We define three key ratios in this section: the **Liquidity Ratio (Liq)**, the **Leverage Ratio (Lev)**, and the **Loan to Debt Ratio (LDR)**.

- (1) The Liquidity Ratio is defined as the ratio of short-term assets (STA) to total liabilities (L)

$$Liq = \frac{\text{Short-Term Assets}}{\text{Total Liabilities}} \quad (2.2.1)$$

- (2) The Leverage Ratio is defined as the ratio of total debt (D) to shareholder funds (SHF)

$$Lev = \frac{\text{Total Debt}}{\text{Shareholder Funds}} \quad (2.2.2)$$

- (3) The Loan to Debt Ratio is defined as the ratio of total loans (General Advances + Advances to Other Banks [GA + ATB]) to total debt (D)

$$LDR = \frac{\text{Total Advances/Loans}}{\text{Total Debt}} \quad (2.2.3)$$

- (4) Of the three ratios, the Liquidity Ratio (Liq) and Leverage Ratio (Lev) are used to restore balance between the asset and liability side following a perturbation, while the Loan to Debt Ratio (LDR) is used, along with additional parameters, to assess the state of the bank.
- (5) Reasonable estimates for the threshold values of the above ratios were determined following a discussion with CA V Krishnakumar.

2.3 State Classification Rules

The state of each bank is determined based on the following penalty-based criteria:

- (1) A penalty is assigned if the Loan to Debt Ratio (LDR) falls below a predefined threshold.
- (2) A penalty is assigned if the Long-Term assets (LTA) of the bank are reduced to zero.
- (3) A penalty is assigned if the Advances to Other Banks (ATB) are reduced to zero.
- (4) The state of the bank is then classified based on the total number of penalties:
 - **Healthy (Green):** No penalties
 - **Crisis (Orange):** One penalty
 - **Default (Red):** Two penalties

2.4 Network Perturbation Rules

Risk Transfer Shock

- (1) This shock is used to simulate the collapse of an initial bank due to risky lending practices. As such, general advances are reduced by a specific shock fraction.
- (2) The resulting loss is first absorbed using standard provisions, which are maintained to account for potential failures arising from non-performing assets.
- (3) Any remaining loss is then absorbed by shareholder funds, which may lead to the bank becoming over-leveraged.
- (4) If the bank becomes over-leveraged, it undergoes deleveraging to meet the maximum allowable leverage limit. The excess debt is accordingly reduced from the balance sheet.

- (5) To restore balance, short-term assets are reduced first (ensuring minimum liquidity is maintained). If short-term assets are insufficient to absorb the required adjustment, callbacks are made in the advances to other banks.

- (6) This reduction is distributed equally across the outgoing interbank exposures of the shocked bank and the edge weight from the shocked bank to each connected bank is reduced accordingly.

- (7) The state of the shocked bank is then evaluated.

- (8) This entire process can be seen in Fig. 2 (a)

Credit Shock

- (1) This shock is used to simulate a market-wide fall in the value of short-term assets. As such, short-term assets are reduced by a specific shock fraction for all the banks in the network one by one.
- (2) The resulting loss is directly absorbed by shareholder funds, which may lead to the bank becoming over-leveraged.
- (3) If the bank becomes over-leveraged, it undergoes deleveraging to meet the maximum allowable leverage limit. The excess debt is accordingly reduced from the balance sheet.
- (4) To restore balance, short-term assets are reduced first, ensuring that minimum liquidity is maintained. If short-term assets are insufficient to absorb the required adjustment, callbacks are made in the advances to other banks.
- (5) This reduction is distributed equally across the outgoing interbank exposures of the shocked bank, and the edge weight from the shocked bank to each connected bank is reduced accordingly.
- (6) The state of each affected bank is then evaluated.
- (7) This entire process can be seen in Fig. 2(b).

Funding Shock

- (1) This shock is used to propagate stress across the interbank network following an initial shock. When Bank A calls back the advances it has made to Bank B, Bank B is said to be experiencing a funding shock.
- (2) When the funding of a bank is reduced, it restores balance by liquidating its long-term assets and recalling advances made to other banks. For simplicity, this adjustment is assumed to occur in equal proportion.
- (3) The realizable value of long-term assets is assumed to depend on the state of the bank. A healthy bank realizes the full value, while a bank in crisis realizes only 75% of the true value.
- (4) The reduction in assets creates an imbalance in the balance sheet, which is absorbed by shareholder funds.
- (5) If the bank becomes over-leveraged, it undergoes deleveraging to meet the maximum allowable leverage limit. The excess debt is accordingly reduced from the balance sheet.
- (6) To restore balance, short-term assets are reduced first, ensuring that minimum liquidity is maintained. If

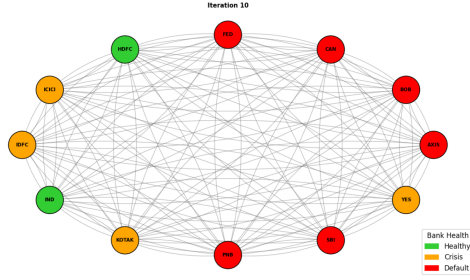


Fig. 3. Punjab National Bank Scam (2018), Final States

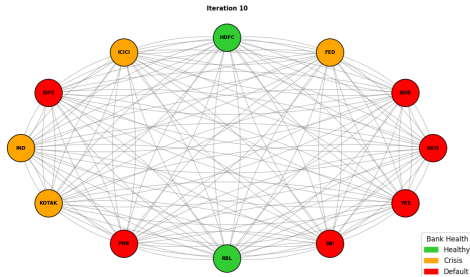


Fig. 4. YES Bank Scam (2020), Final States

short-term assets are insufficient to absorb the required adjustment, callbacks are made in the advances to other banks.

- (7) The state of each bank is then calculated based on the updated balance sheet.
- (8) Once all the banks have had a funding shock executed, this process is continued iteratively for four or 5 cycles.
- (9) This process can be seen in Fig. 2(c).

3. RESULTS AND DISCUSSION

3.1 Punjab National Bank Crisis (2018)

As shown in Table C.1 and Fig.3, our model predicts that during the 2018 crisis, Punjab National Bank, Canara Bank, Bank of Baroda, Axis Bank, State Bank of India, and Federal Bank were among the banks at high risk of defaulting.

Official reports and newspaper articles indicate that the banks affected by the scam included Punjab National Bank, Canara Bank, Axis Bank, and State Bank of India [4, 5]. This indicates that for the most part our model is able to accurately identify banks that are at high risk of defaulting, provided they fall within the scope of the model.

The model accurately identifies several affected banks; however, it appears to overestimate the severity of impact for some institutions.

3.2 YES Bank Crisis (2020)

As shown in Table C.2 and Fig.4, our model predicts that during the 2020 crisis, Punjab National Bank, Bank of

Baroda, Axis Bank, State Bank of India, IDFC Bank, and YES Bank were among the banks at high risk of defaulting.

In 2020 both the effects of both YES Bank crisis (risk transfer shock) and the COVID-19 pandemic (credit shock) were felt. Consequently, most reports are unable to clearly attribute banking system failures to a single trigger.

However, since both shocks were simulated in our model, we rely on reports/articles assessing the overall condition of the banking system in 2020 for validating the model.

From these reports we gather that PNB, SBI, Central Bank of India, Bank of Baroda, and Canara Bank were the public sector banks most significantly affected, while Axis Bank and IDBI Bank were among the impacted private sector banks; however, the extent of damage remains uncertain (different sources cite different figures) [6,7].

3.3 Model Drawbacks

As pointed out earlier, the health of banks outside the scope of our model (that is, banks not included in the BANKNIFTY index) cannot be predicted using this framework.

For example, during the 2018 crisis, Allahabad Bank and Union Bank of India were also severely impacted by the PNB crisis, but their default risk is not captured by our model. A similar case arises when analyzing the 2020 crisis, where IDBI Bank and several other smaller banks experienced severe financial stress, but these effects are not picked up by our model.

Additionally, any secondary effects that arise from the defaulting of these banks is also not captured by this model.

Besides this, the current model also appears to overestimate the extent of damage to certain banks (particularly smaller banks such as Federal Bank and Bank of Baroda). This is primarily due to the disproportionate ratio between their BANKNIFTY index weights and their total asset or liability values. As a result, exposures to affected banks may be overestimated, leading to inaccuracies in predictions.

4. CONCLUSIONS

It is imperative to keep in mind the drawbacks mentioned above while interpreting the outputs of this model. While the model is able to predict the failure of large banks with reasonable accuracy. For more holistic assessments of the banking system, using this model in conjunction with a model of wider scope (such as the framework proposed by Jorge A. Chan-Lau) can provide vastly more accurate insights.

5. ACKNOWLEDGEMENT

We thank CA V Krishnakumar, for guiding us in the construction of the model and choice of parameters. We are grateful to Dr Karthik Raman for providing us with the opportunity to undertake this project as part of this course.

6. REFERENCES

- (1) Jorge A Chan-Lau, *Balance Sheet Network Analysis of Too-Connected-to-Fail Risk in Global and Domestic Banking Systems*, IMF Working Paper, 2010.
 - (2) Christian Upper and Andreas Worms, *Estimating bilateral exposures in the German interbank market: Is there a danger of contagion?*, European Economic Review, Volume 48, pages 827-849, 2004.
 - (3) Simon Wells, *Financial interlinkages in the United Kingdom's interbank market and the risk of contagion*, Working Paper no. 230 (Bank of England), 2004.
 - (4) Wikipedia - Punjab National Bank Scam (2018)
https://en.wikipedia.org/wiki/Punjab_National_Bank_Scam
 - (5) Times of India, *PNB fraud affects these banks too*, 2018.
 - (6) From Trust to Turmoil—The Yes Bank Financial Scam
<https://ifsa-network.com/publications/from-trust-to-turmoil-the-yes-bank-financial-scam/>
 - (7) Financial Express, *PNB, SBI among worst hit PSU-banks by bad loans to top 50 wilful defaulters*, 2020
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7. SOURCES OF DATA

- a) <https://www.nseindia.com/>
- b) <https://www.tijorifinance.com/dashboard/>

PDF-scrapper was used on [a] to obtain Schedule specific data from the Annual Reports. Web-scrapper was used on [b] to obtain Balance sheet data.

Appendix A. ONTOLOGY

- i) **Assets:** The recorded resources of economic value (Cash, Gold, Government bonds etc.) that a bank owns or controls.
- ii) **Short Term Assets:** The sum of the bank's cash reserves held with the RBI and money at call and short notice (obtained from the sum of totals of Schedules 6 and 7).
- iii) **Long Term Assets:** The sum of the bank's investments, fixed assets, and other assets (obtained from the sum of totals of Schedules 9, 10, and 11).
- iv) **Advances to Other Banks:** The sum of a bank's advances to other banks in India and syndicated loans to banks in India (Obtained from Schedule 12).
- v) **General Advances:** All other advances not included in (iv) (obtained by subtracting (iv) from total advances).
- vi) **Liabilities:** The financial obligations and debts a bank owes, primarily to customers, shareholders, and other financial institutions.
- vii) **Shareholder Funds:** The sum of the bank's capital and its reserves and surplus (obtained from the sum of totals of Schedules 1 and 2).
- viii) **Non-Performing Assets:** Non-performing assets (NPAs) are loans or advances where borrowers fail to make interest or principal payments (advances or loans that have gone stale).
- ix) **Standard Provisions:** The money a bank sets aside to cover potential loan repayments (Obtained from Schedule 5 in provisions against standard assets)
- x) **Other Provisions:** All other provisions not included in (ix) (obtained by subtracting (ix) from total provisions).
- xi) **Debt:** The sum of a bank's borrowing and the total deposits it holds (obtained from the sum of totals of Schedules 3 and 4).

Appendix B. USAGE OF AI

We would like to begin this section by stating that both authors of this paper, along with an expert in this domain, were actively involved in understanding the system, simplifying banks for a financial network, building and specifically

designing a shock mechanism for Indian networks, and establishing rules and states to determine the health of the financial network and its constituent banks during a financial crisis.

The implementation of this system involves multiple phases: (i) Data Extraction and Preprocessing, (ii) Class Bank and Network Perturbation, and (iii) Visualization and Debugging. AI was used solely to aid and assist in finding errors in the codebase and in writing boilerplate code, and only in phases (i) and (iii). AI was in no way used to write or help write the content of this report, nor to verify the report for mistakes. Grammarly was used to find grammatical errors and improve sentence structure.

The use of AI was specifically to aid in the design of website scrapers and PDF parsers. It was extremely difficult to go through multiple existing libraries and familiarize oneself with their syntax; hence, AI was used to assist with XML and HTML syntax. It was also used to define functions that would overlay bank snapshots after each perturbation to create GIFs, and to visualize the results in a cleaner and more comprehensive manner.

Code to display the interbank exposure network cleanly:

```
nx.draw_networkx_edges(
    network,
    pos,
    edgelist=list(network.edges()),
    arrows=True,
    arrowstyle='->',          # triangular arrowhead
    arrowsize=40,            # bigger arrow head
    width=1.2,               # thicker edges
    edge_color="dimgray",
    alpha=0.85,
    min_source_margin=15,    # prevents arrows touching nodes
    min_target_margin=15,
    connectionstyle="arc3,rad=0.12" # separates bidirectional edges
)

edge_labels_scaled = {
    (u,v): f"{w/1e7:.3f}"
    for (u,v), w in edge_labels.items()
}
```

Used Claude to debug an issue in the regulatory ratio implementation logic:

```
if old_state != self.health:
    #imp debug, see which bank causes staleness
    print(
        f"\nSTATE CHANGE: {self.name}: {old_state} -----> {self.health} | "
        f"LDR={ldr:.3f} | LIQ={liq:.3f} | LEV={lev:.2f} | "
        f"ATB={self.atb:.2e} | LTA={self.lta:.2e} | SHF={self.shf:.2e}"
    )

#debug statement, based on error one iteration causes all banks to go into crisis and be stale
print(f"\n[DELEVERAGE START] {self.name} | leverage={self.leverage():.2f} | "
      f"ATB={self.atb:.2e} | SHF={self.shf:.2e}")

# SHF gone negative; To change that bank is insolvent, ignoring deleverage
if self.shf <= 0:
    #add statement to imply why shf <0 implies bank going default
    print(f"[INSOLVENT] {self.name} | SHF={self.shf:.2e} forcing Default")
    self.health = "Default"
    self.penalties = 2
    return 0.0

print(
    #debug statement
    f"[DELEVERAGE PROP] {self.name} -----> {neighbour} | "
    f"share={share:.2e} (w={weight:.2e}) | ATB: {before:.2e} → {after:.2e}"
)
```

```

)

if before > 0 and after == 0:
    #debug statement
    print(f"[ATB ZERO FLAG] {neighbour} -----> ATB={after:.2e}")

print("debug for ratio limits")
print(f"{name:<8} | LDR={ldr:.3f} | LIQ={liq:.3f} | LEV={lev:.3f} | ATB={bank.atb:.2e} |
      LTA={bank.lta:.2e} | penalties={penalties}")

print(f"ITER Network Snapshot")

for b in banks.values():
    print(
        f"{b.name:<6} | "
        f"{b.health:<8} | "
        f"LDR={b.ldr():.2f} | "
        f"LIQ={b.liquidity():.2f} | "
        f"LEV={b.leverage():.2f} | "
        f"ATB={b.atb:.1e}"
    )

```

Generating a GIF illustrating the evolution of bank states across successive iterations:

```

fig_gif, ax_gif = plt.subplots(figsize=(13, 7))
def update(frame):
    ax_gif.clear()
    draw_panel(ax_gif, snapshots[frame], snapshot_titles[frame], pnb_network, pos)
    ax_gif.legend(handles=legend_patches, loc="lower right",
                  fontsize=10, title="Bank Health", framealpha=0.9)

def record_snapshot(banks, label):
    snapshots.append({name: bank.health for name, bank in banks.items()})
    snapshot_titles.append(label)
record_snapshot(banks, "Stage 0: Baseline")

banks["YES"].credit_shock(0.30, pnb_network, banks)
record_snapshot(banks, "Stage 1: YES Bank Credit Shock (30%)")
total = len(snapshots)

indices = sorted(set(
    [0, 1] + [2 + round(i * (total - 3) / 4) for i in range(5)]
))[:7]

selected_snapshots = [snapshots[i] for i in indices]
selected_titles = [snapshot_titles[i] for i in indices]

def draw_panel(ax, snap, title, network, pos):
    colors = [health_color[snap[node]] for node in network.nodes()]
    nx.draw_networkx_nodes(network, pos, node_color=colors, edgecolors="black",
                           node_size=1600, ax=ax)
    nx.draw_networkx_labels(network, pos, font_size=7, font_weight="bold", ax=ax)
    nx.draw_networkx_edges(network, pos, arrows=True, arrowstyle="->", arrowsize=12,
                           width=0.7, edge_color="dimgray", alpha=0.55, min_source_margin=10,
                           min_target_margin=10, connectionstyle="arc3,rad=0.12", ax=ax)
    ax.set_title(title, fontsize=9, fontweight="bold", pad=6)
    ax.axis("off")

ani = animation.FuncAnimation(
    fig_gif, update,
    frames=len(snapshots),
    interval=800,
    repeat=True)

```


Appendix C. TABLES

Table C.1. Final Iteration Snapshot: Punjab National Bank Crisis (2018)

Bank	State	LDR	Liquidity	Leverage	Advances to Other Banks(INR)
AXIS	Default	0.71	0.08	11.22	3.5×10^{10}
BOB	Default	0.59	0.22	21.65	1.2×10^{11}
CAN	Default	0.63	0.10	18.16	4.8×10^{10}
FED	Default	0.71	0.07	12.13	4.6×10^9
HDFC	Healthy	0.77	0.06	8.02	2.7×10^{10}
ICICI	Crisis	0.70	0.10	8.89	3.2×10^{11}
IDFC	Crisis	0.55	0.05	6.20	9.8×10^8
IND	Healthy	0.76	0.10	7.22	4.8×10^6
KOTAK	Crisis	0.74	0.09	5.85	8.0×10^9
PNB	Default	0.51	0.13	∞	1.5×10^{10}
SBI	Default	0.65	0.07	18.66	6.1×10^{11}
YES	Crisis	0.72	0.09	9.87	3.6×10^{10}

Table C.2. Final Iteration Snapshot: YES Bank Crisis (2020)

Bank	State	LDR	Liquidity	Leverage	Advances to Other Banks (INR)
AXIS	Default	0.69	0.09	12.55	5.2×10^{10}
BOB	Default	0.65	0.12	22.24	1.5×10^{11}
FED	Crisis	0.76	0.06	11.95	1.4×10^{10}
HDFC	Healthy	0.78	0.07	6.97	3.9×10^{10}
ICICI	Crisis	0.71	0.08	7.57	3.8×10^9
IDFC	Default	0.59	0.06	11.41	6.6×10^{10}
IND	Crisis	0.77	0.05	9.08	0.0×10^0
KOTAK	Crisis	0.80	0.08	6.02	0.0×10^0
PNB	Default	0.64	0.10	16.68	2.1×10^{10}
RBL	Healthy	0.77	0.08	9.30	9.0×10^8
SBI	Default	0.64	0.06	24.21	9.5×10^{11}
YES	Default	0.81	0.05	15.00	0.0×10^0